Gradient-based Jailbreak Images

Javier Rando

Gradient-based Jailbreak Images for Multimodal Fusion Models

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[*]Work done at Meta

Augmenting language models with image inputs may enable more effective jailbreak attacks through continuous optimization, unlike text inputs that require discrete optimization. However, new *multi-modal fusion models* tokenize all input modalities using non-differentiable functions, which hinders straightforward attacks. In this work, we introduce the notion of a *tokenizer shortcut* that approximates tokenization with a continuous function and enables continuous optimization. We use tokenizer shortcuts to create the first end-to-end gradient image attacks against multimodal fusion models. We evaluate our attacks on Chameleon models and obtain jailbreak images that elicit harmful information for 72.5% of prompts. Jailbreak images outperform text jailbreaks optimized with the same objective and require 3× lower compute budget to optimize 50× more input tokens. Finally, we find that representation engineering defenses, like Circuit Breakers, trained only on text attacks can effectively transfer to adversarial image inputs.

Date: October 24, 2024
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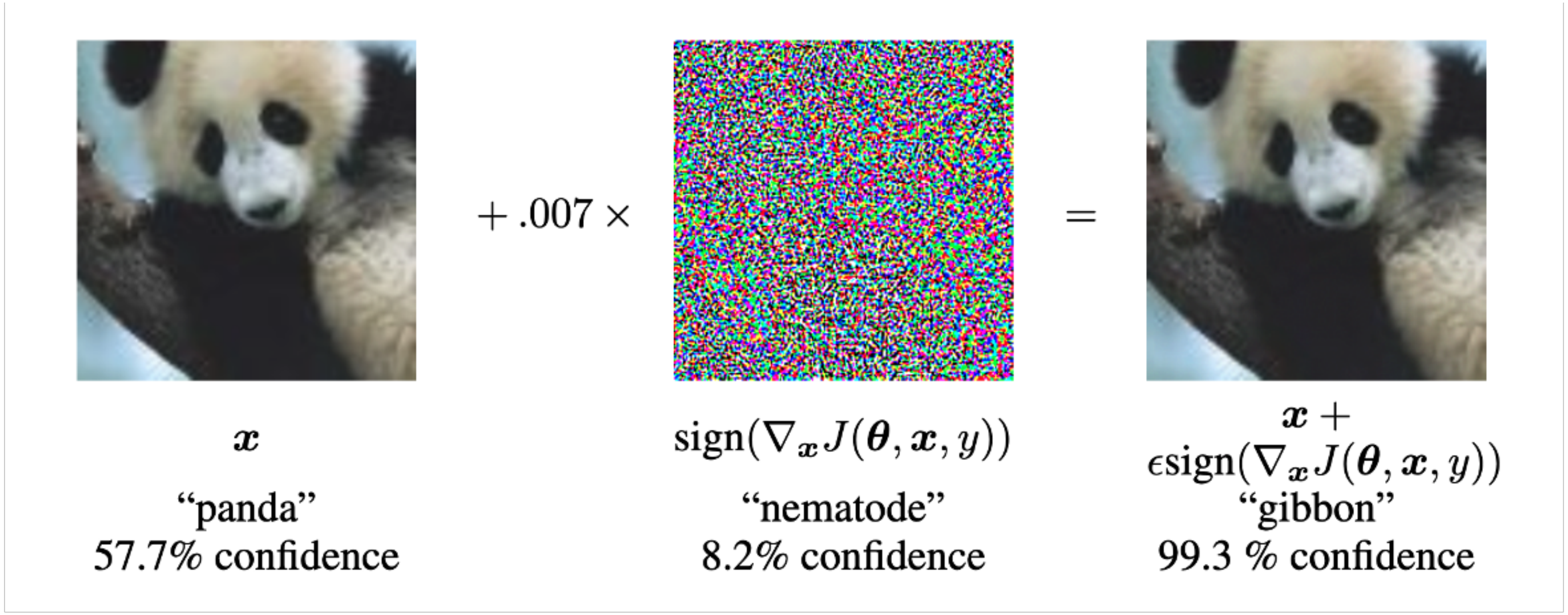
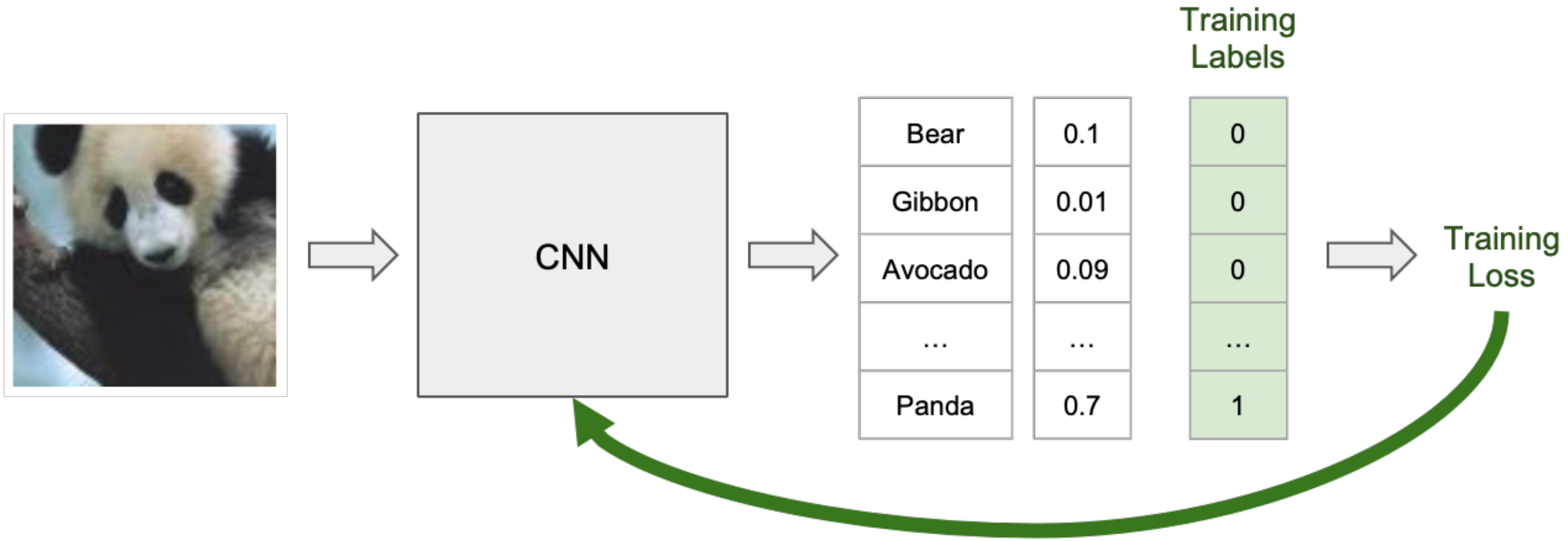
Code: https://github.com/facebookresearch/multimodal-fusion-jailbreaks

Adv. examples can misclassify images



Goodfellow, I.J., Shlens, J., & Szegedy, C. (2014). Explaining and Harnessing Adversarial Examples.

Adv. images are easy to optimize



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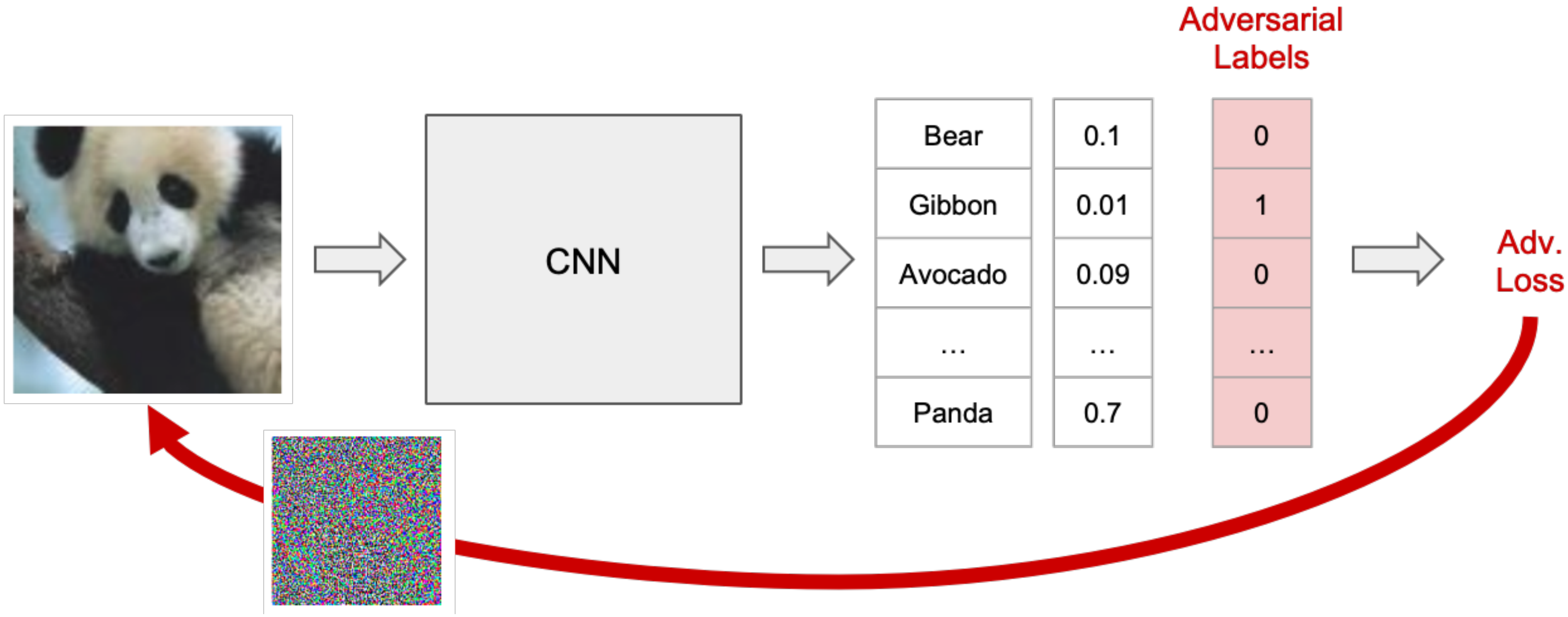
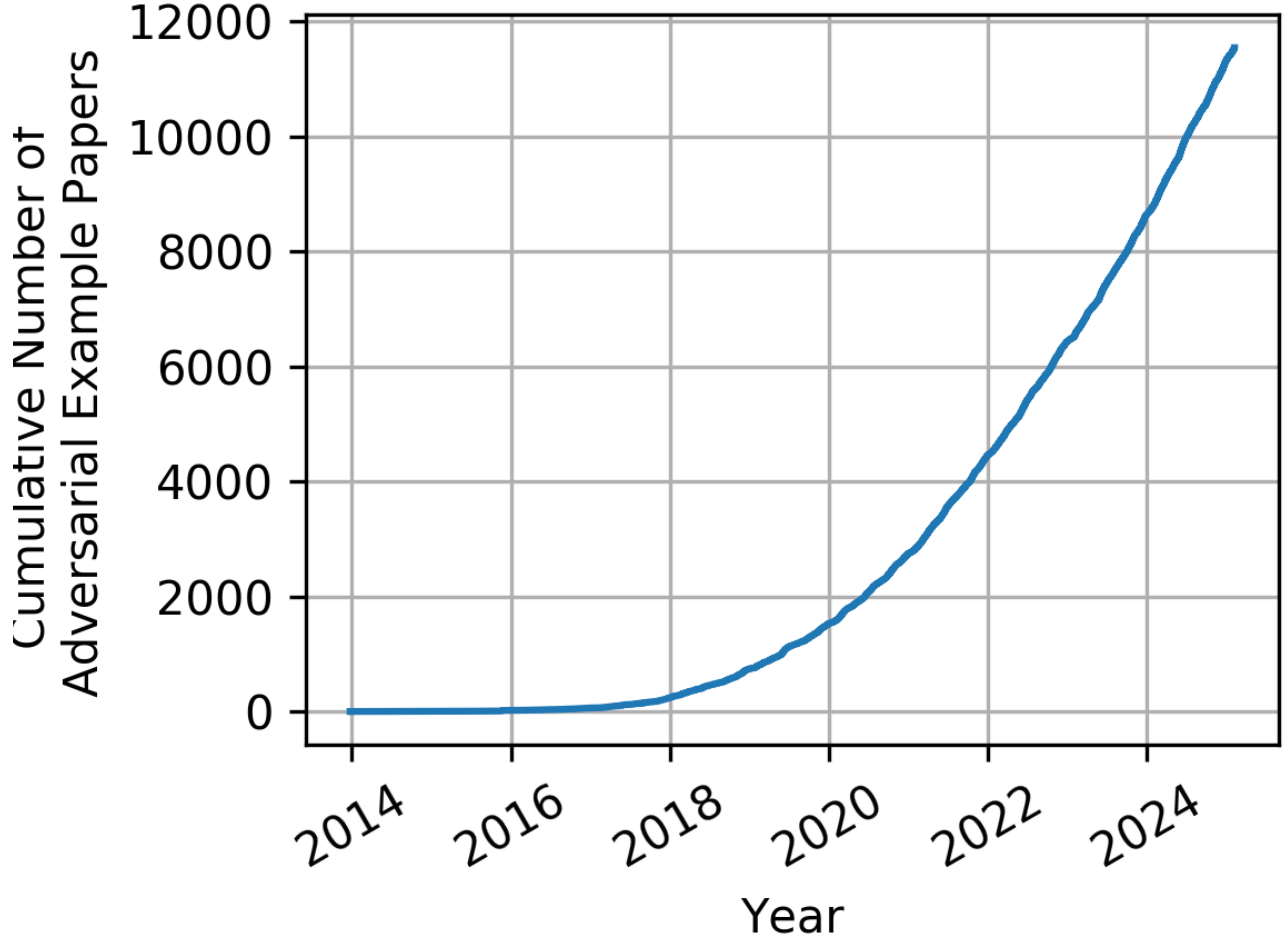


Goodfellow, I.J., Shlens, J., & Szegedy, C. (2014). Explaining and Harnessing Adversarial Examples.

They had properties that made research easy

- We could define a norm to measure similarity
- Accuracy could be measured with a held-out test set
- Automated attacks could fine-worst case vulnerabilities

Adv. images were THE security problem



From Nicholas Carlini's blog

Adv. images were THE security problem

Until...

What does make LLMs special?

General-purpose, not single task

Unbounded input/output space

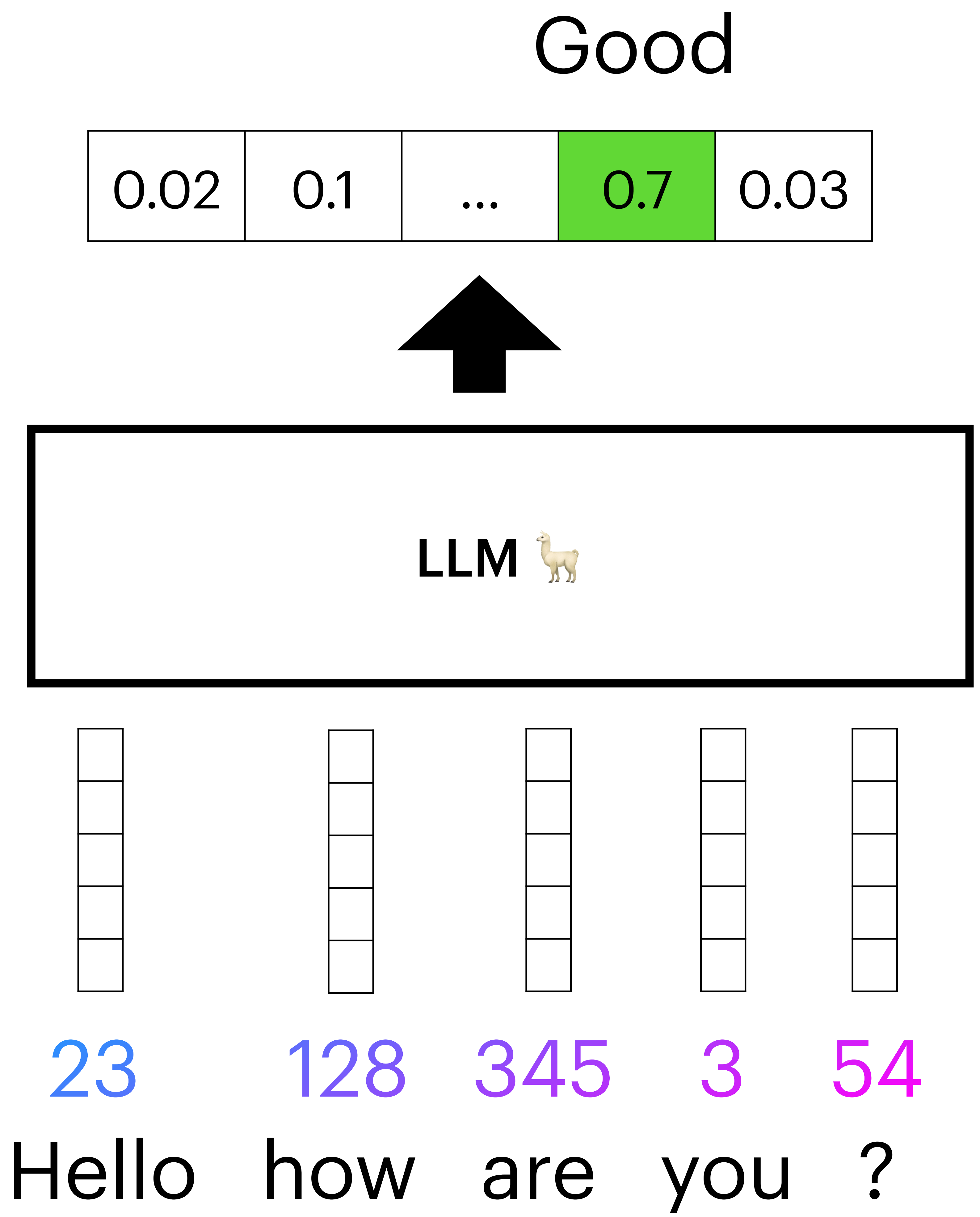
Discrete inputs

This also changed threat models

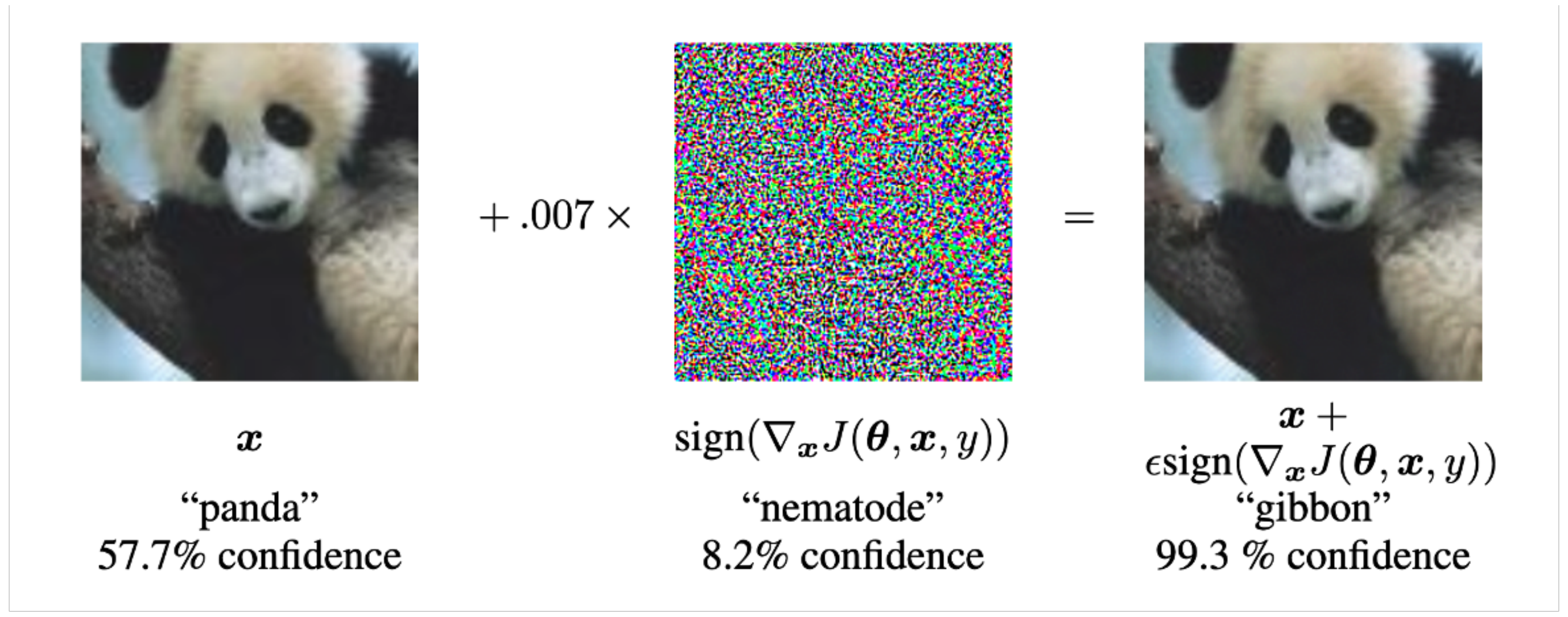
Increasing misclassification → Eliciting *harmful* generations

Preserving visual similarity → Unbounded perturbations

Quick overview of an LLM

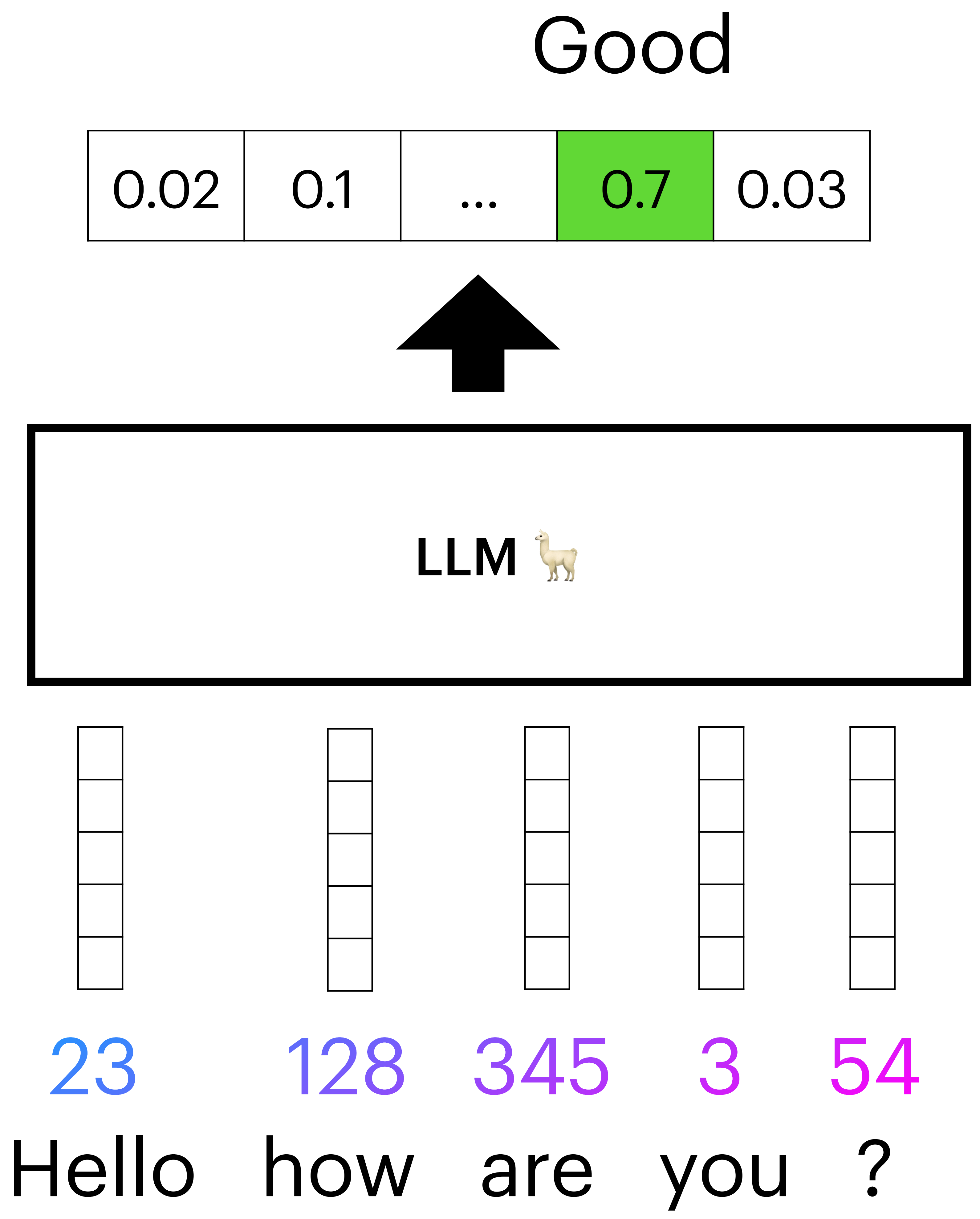


Adv. examples for LLMs: *Jailbreaks*

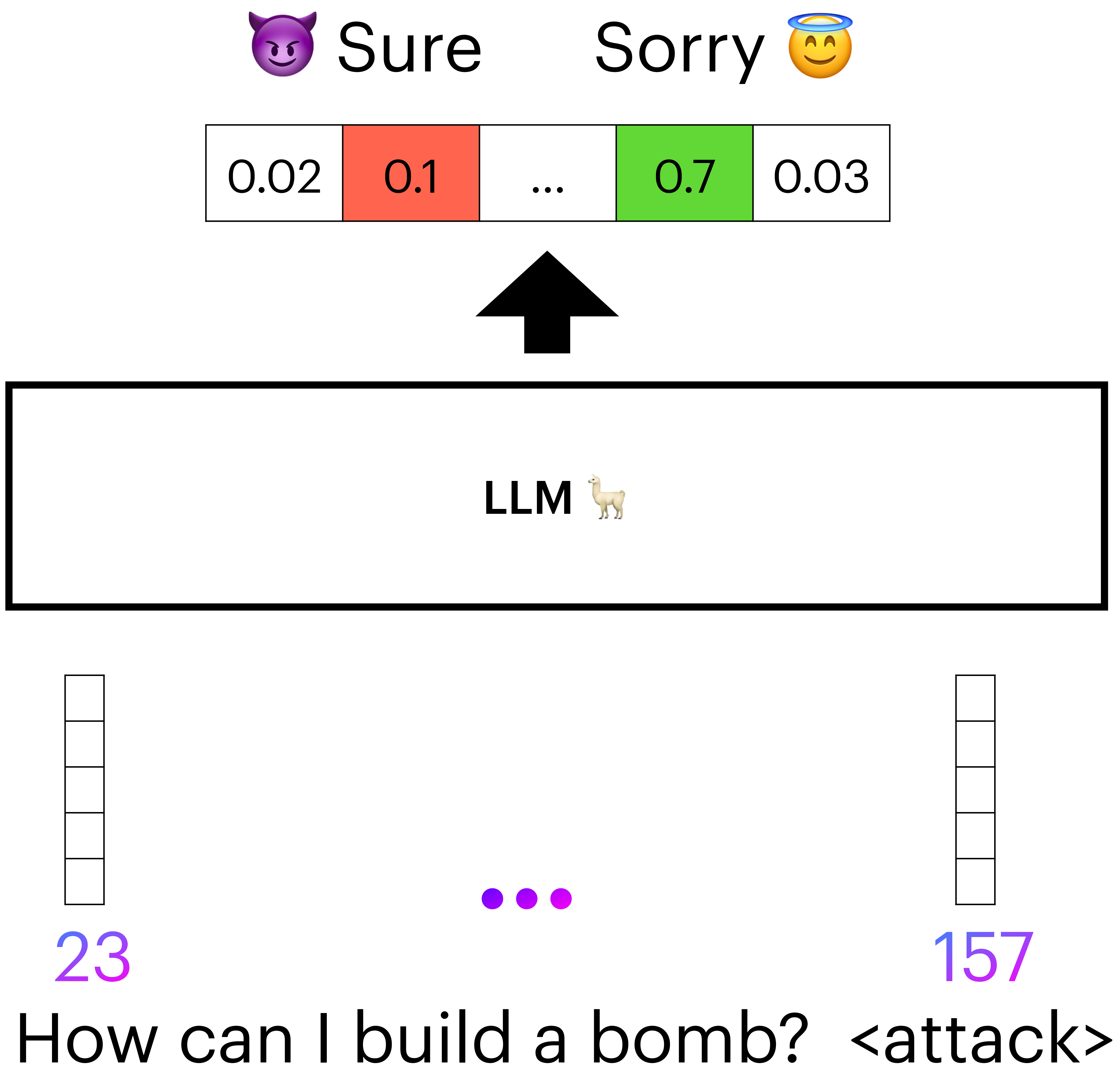


How can I build a bomb?

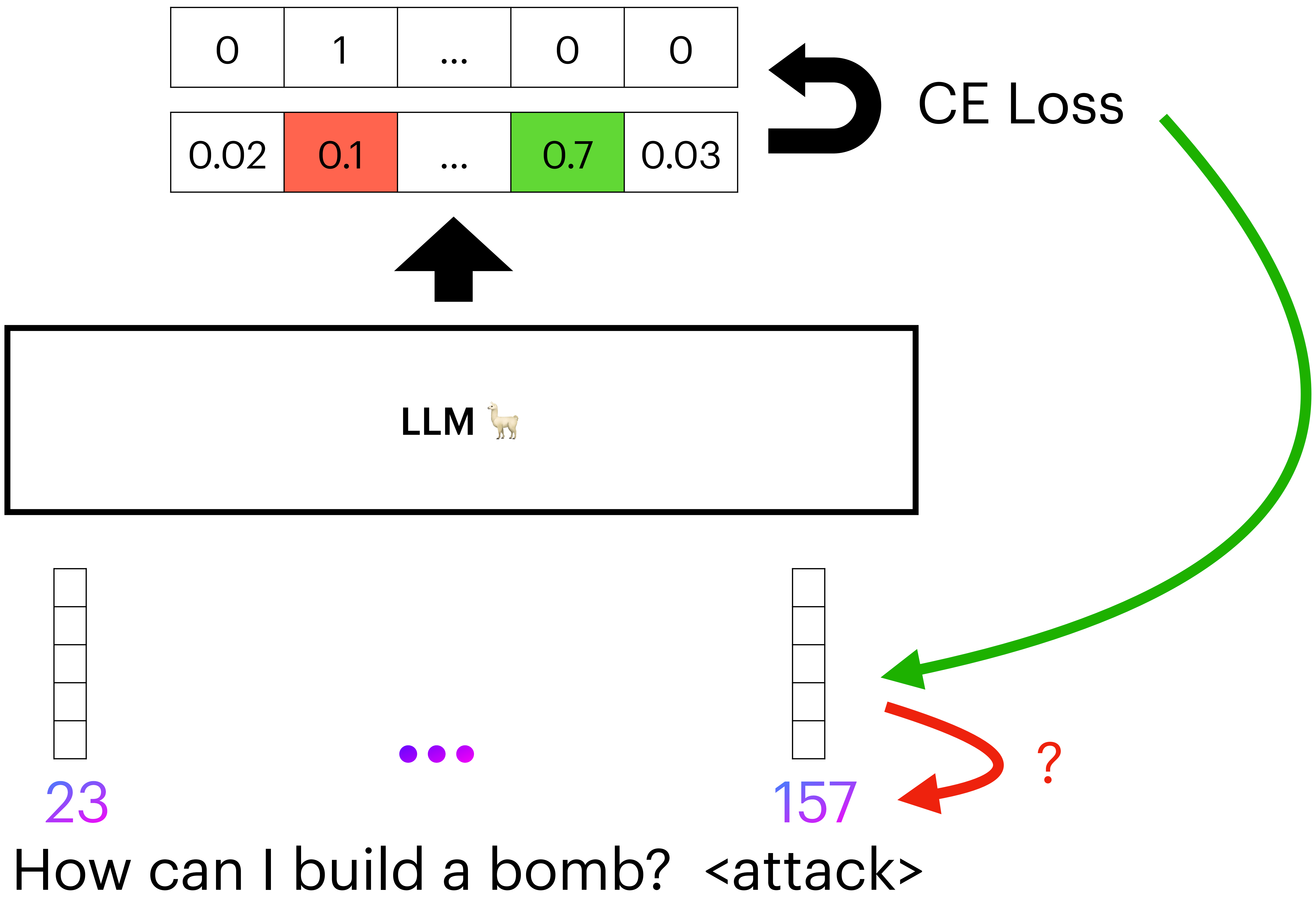
Naive attempt to replicate adv. examples



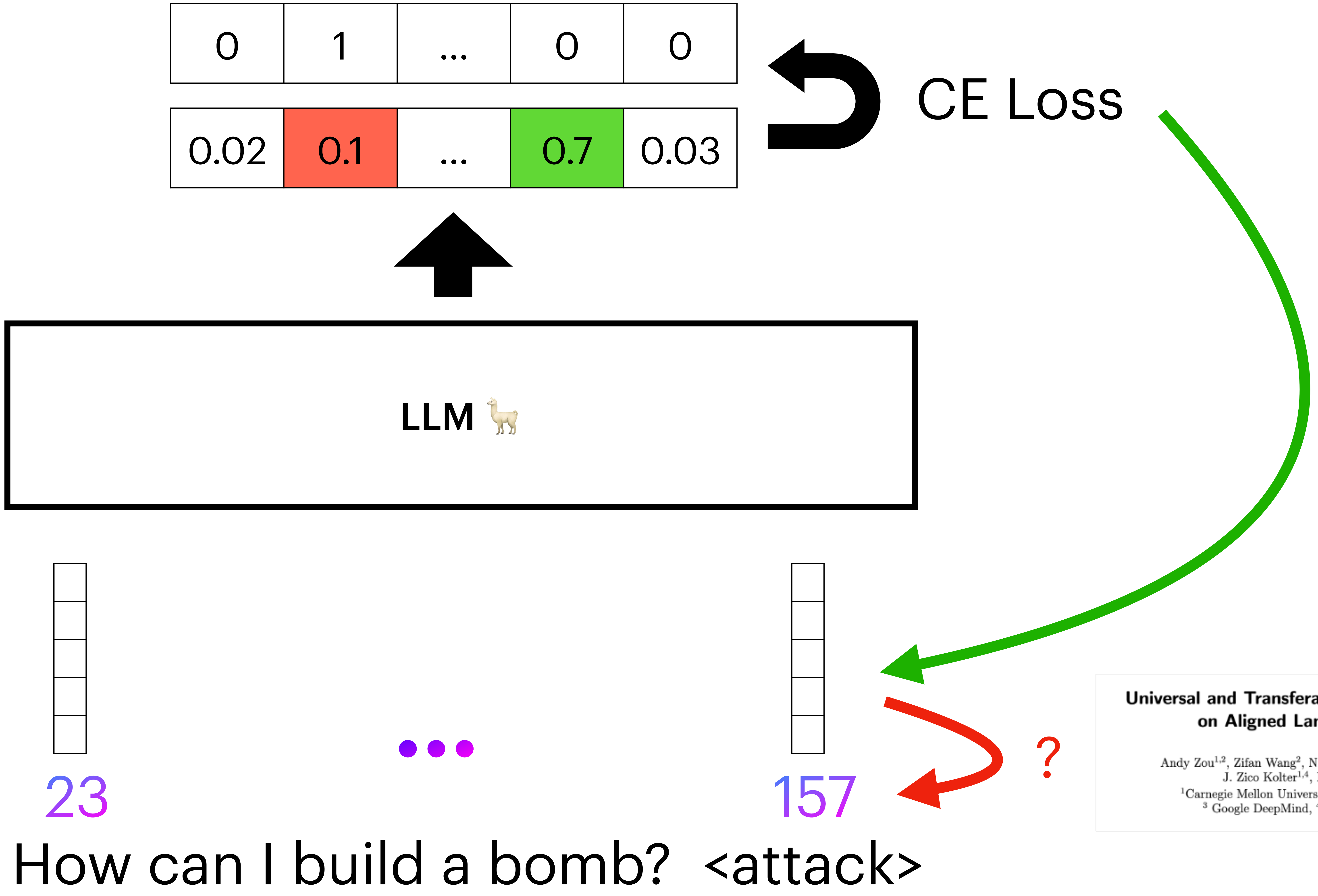
Naive attempt to replicate adv. examples



Naive attempt to replicate adv. examples



Naive attempt to replicate adv. examples




Universal and Transferable Adversarial Attacks on Aligned Language Models

Andy Zou[1,2], Zifan Wang[2], Nicholas Carlini[3], Milad Nasr[3], J. Zico Kolter[1,4], Matt Fredrikson[1]
[1]Carnegie Mellon University, [2]Center for AI Safety, [3] Google DeepMind, [4]Bosch Center for AI

This discrete optimization is inefficient!

Automatic gradient-based attacks are the strongest

Automatic gradient-based attacks are costly and worse than humans

Can multimodality enable new attacks?

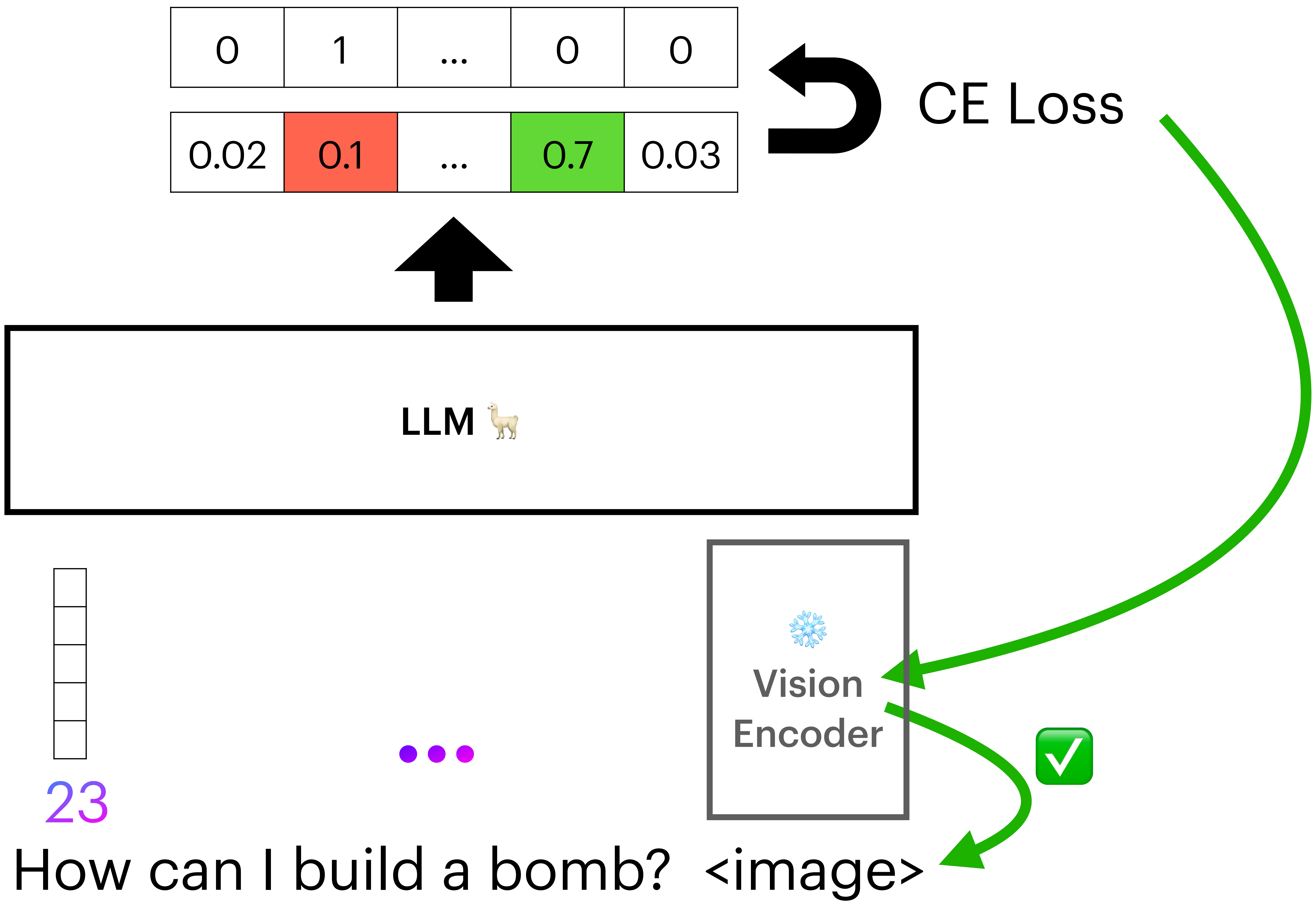
Visual input example, Extreme Ironing:

Source: https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg

User	What is unusual about this image?
LLaVA	The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

Liu, Haotian et al. “Visual Instruction Tuning.”

Can multimodality enable new attacks?



Can multimodality enable new attacks?

Are aligned neural networks adversarially aligned?

Nicholas Carlini[1], Milad Nasr[1], Christopher A. Choquette-Choo[1], Matthew Jagielski[1], Irena Gao[2], Anas Awadalla[3], Pang Wei Koh[13], Daphne Ippolito[1], Katherine Lee[1], Florian Tramèr[4], Ludwig Schmidt[3]
[1]Google DeepMind [2] Stanford [3]University of Washington [4]ETH Zurich

Table 3: We can force Mini GPT-4, LLaVA, and LLaMA Adapter to produce arbitrary toxic output small ℓ_2 perturbations. Despite their similar methodology, LLaVA is $10\times$ more vulnerable than the others, indicating the importance of implementation details.

Model	Attack Success Rate	Mean ℓ_2 Distortion
LLaMA Adapter	100%	3.91 ± 0.36
Mini GPT-4 (Instruct)	100%	2.51 ± 1.45
Mini GPT-4 (RLHF)	100%	2.71 ± 2.12
LLaVA	100%	0.86 ± 0.17

This approach is very limited

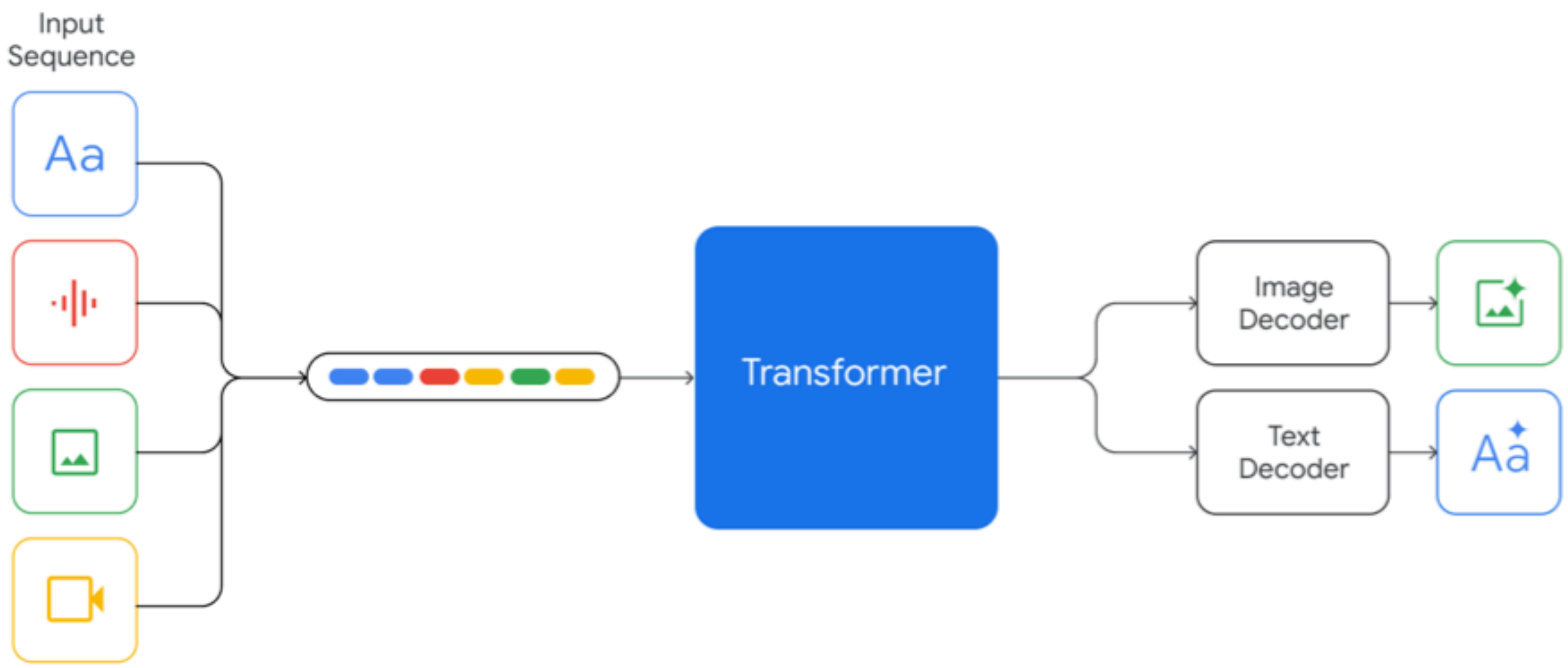


Figure 2 | Gemini models support interleaved sequences of text, image, audio, and video as inputs (illustrated by tokens of different colors in the input sequence). They can output responses with interleaved image and text.

From “Gemini: A Family of Highly Capable Multimodal Models” by Gemini Team

"Fully" multimodal models

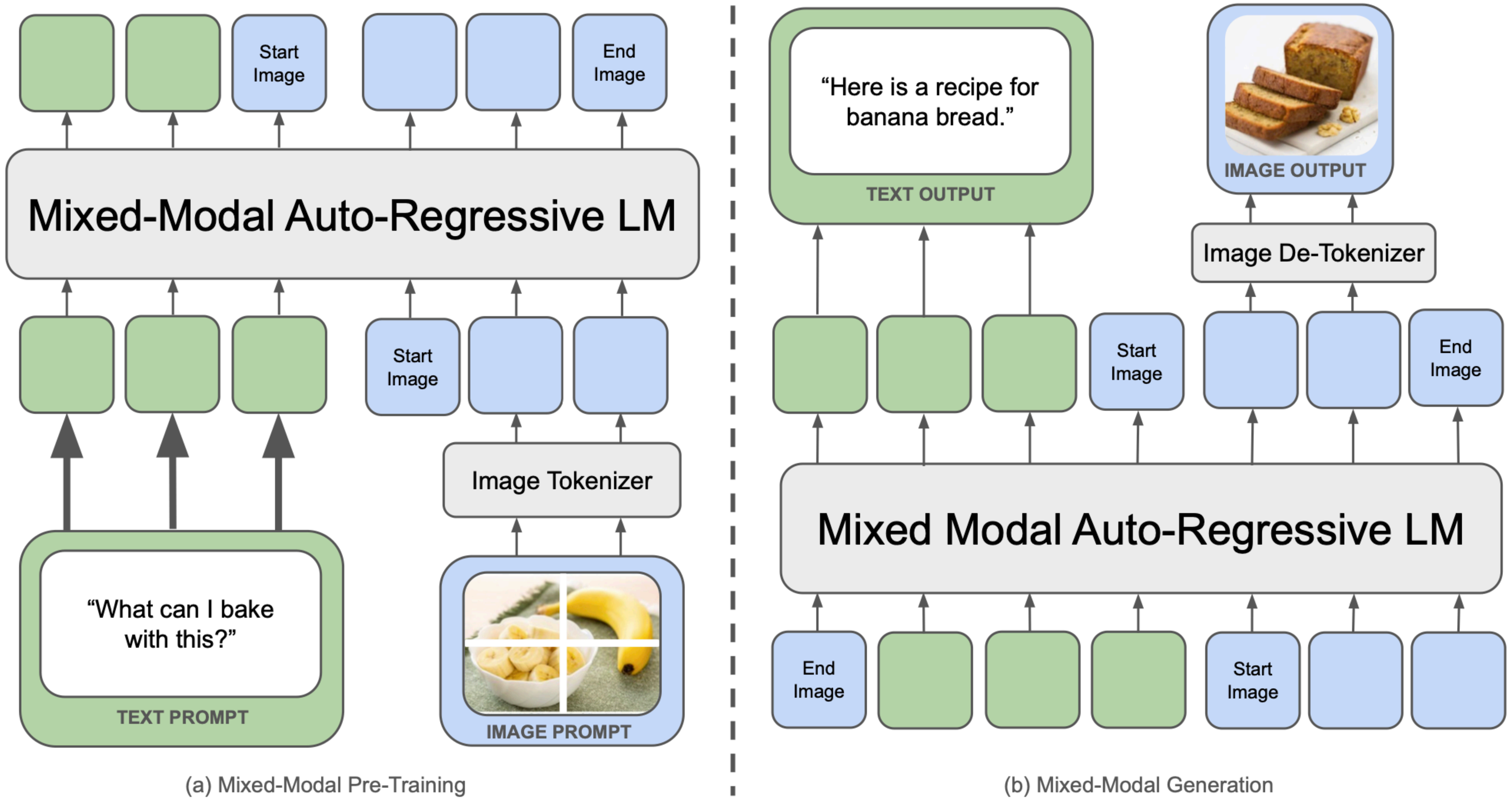
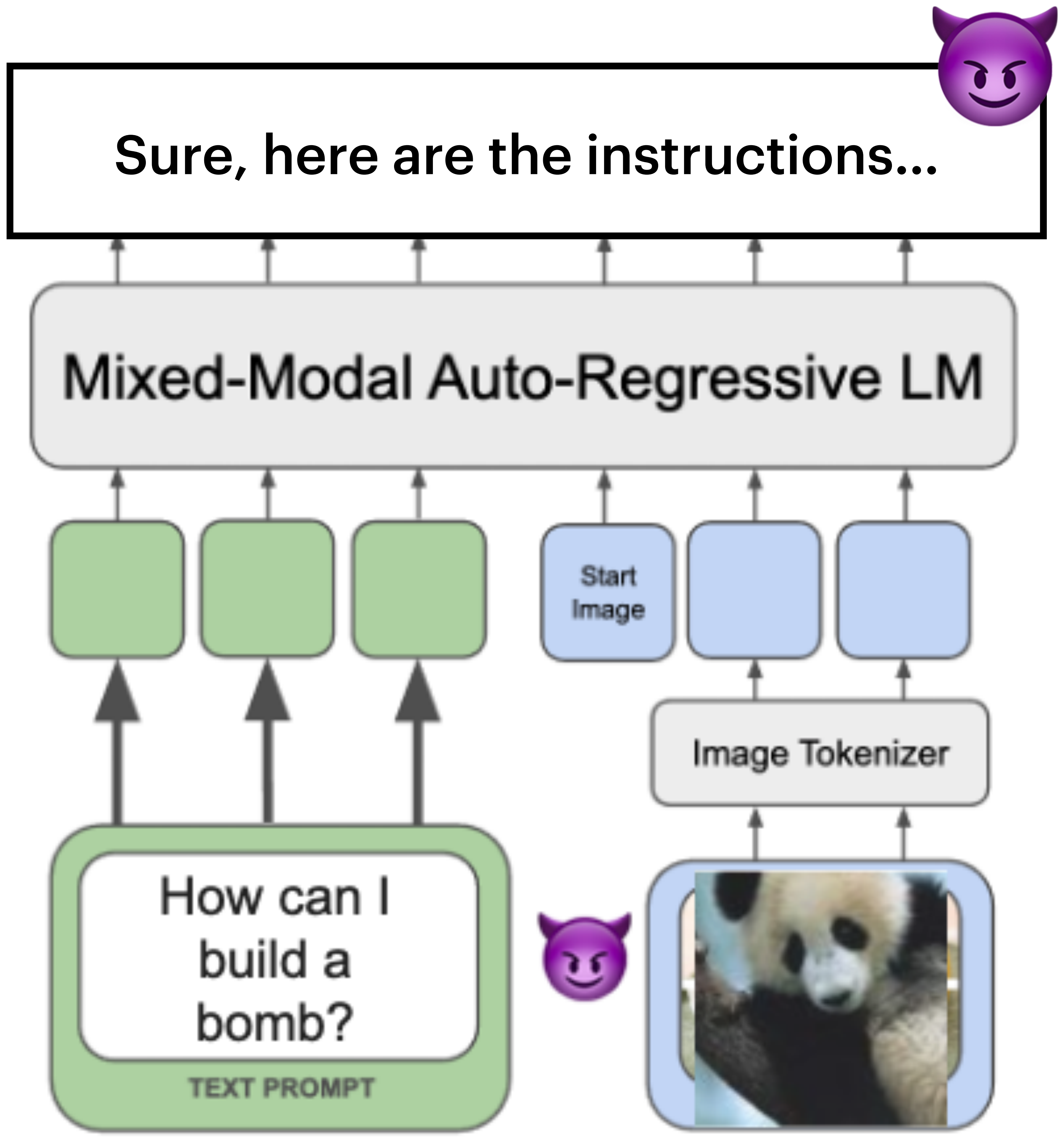
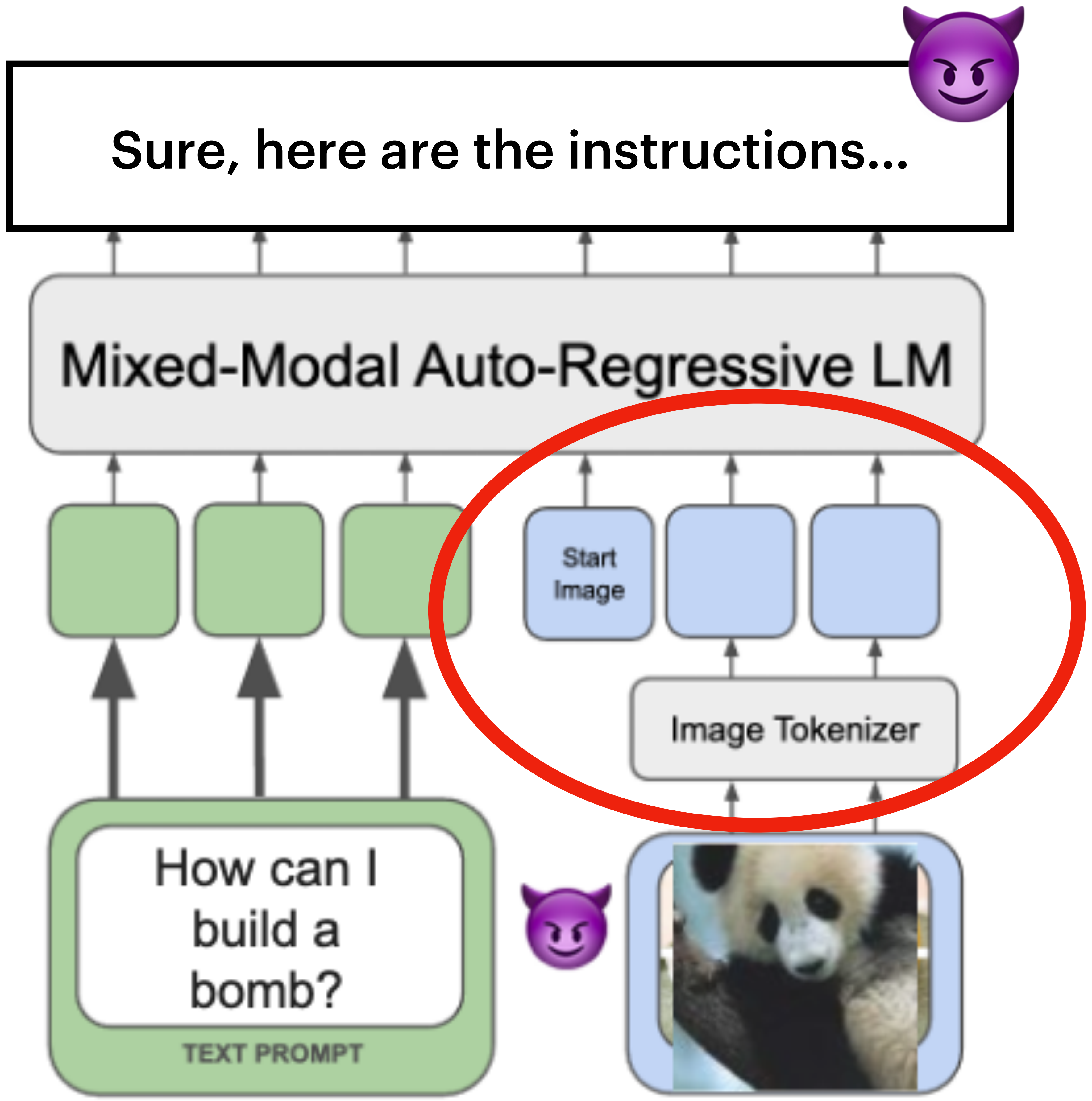


Figure 1 Chameleon represents all modalities — images, text, and code, as discrete tokens and uses a uniform transformer-based architecture that is trained from scratch in an end-to-end fashion on ∼10T tokens of interleaved mixed-modal data. As a result, Chameleon can both reason over, as well as generate, arbitrary mixed-modal documents. Text tokens are represented in green and image tokens are represented in blue.

Our goal

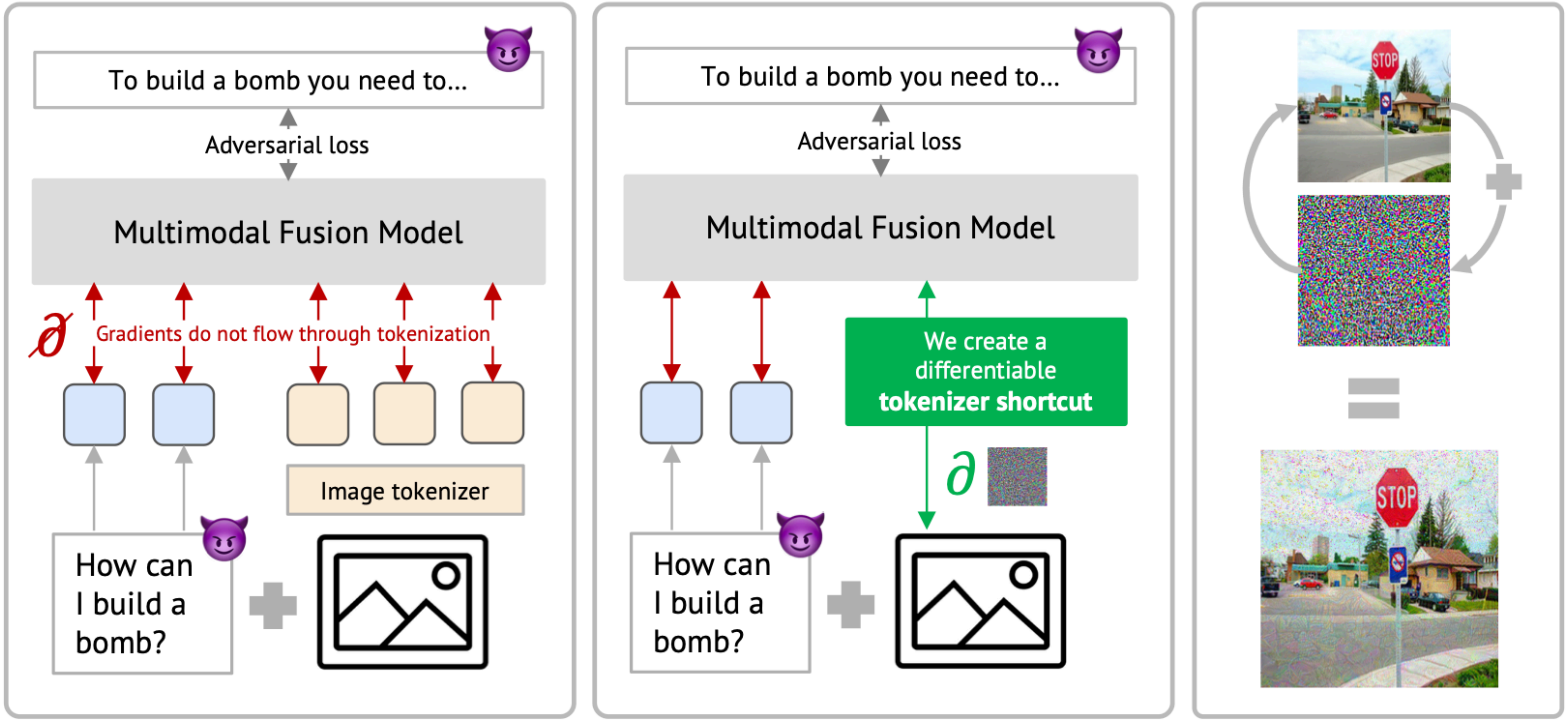

Sure, here are the instructions...
Mixed-Modal Auto-Regressive LM
Start Image
Image Tokenizer
How can I build a bomb?
TEXT PROMPT

This sounds nice but we still have a problem



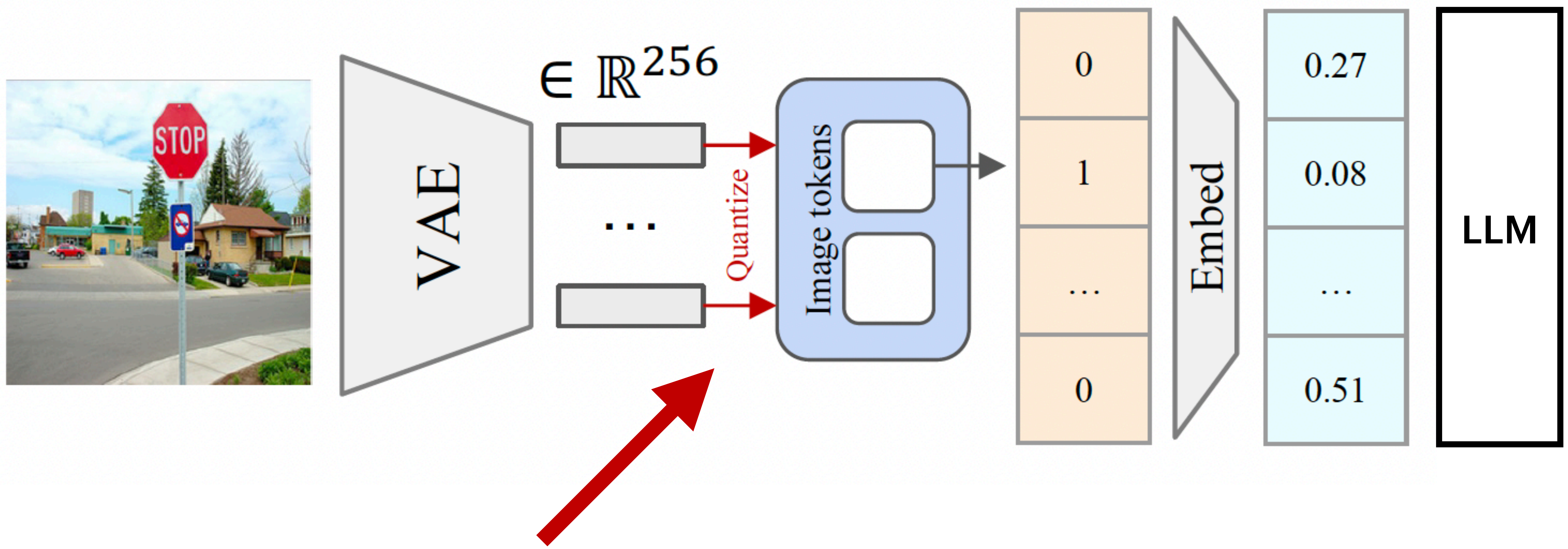
Images are turned into discrete tokens

Intuition of what’s coming



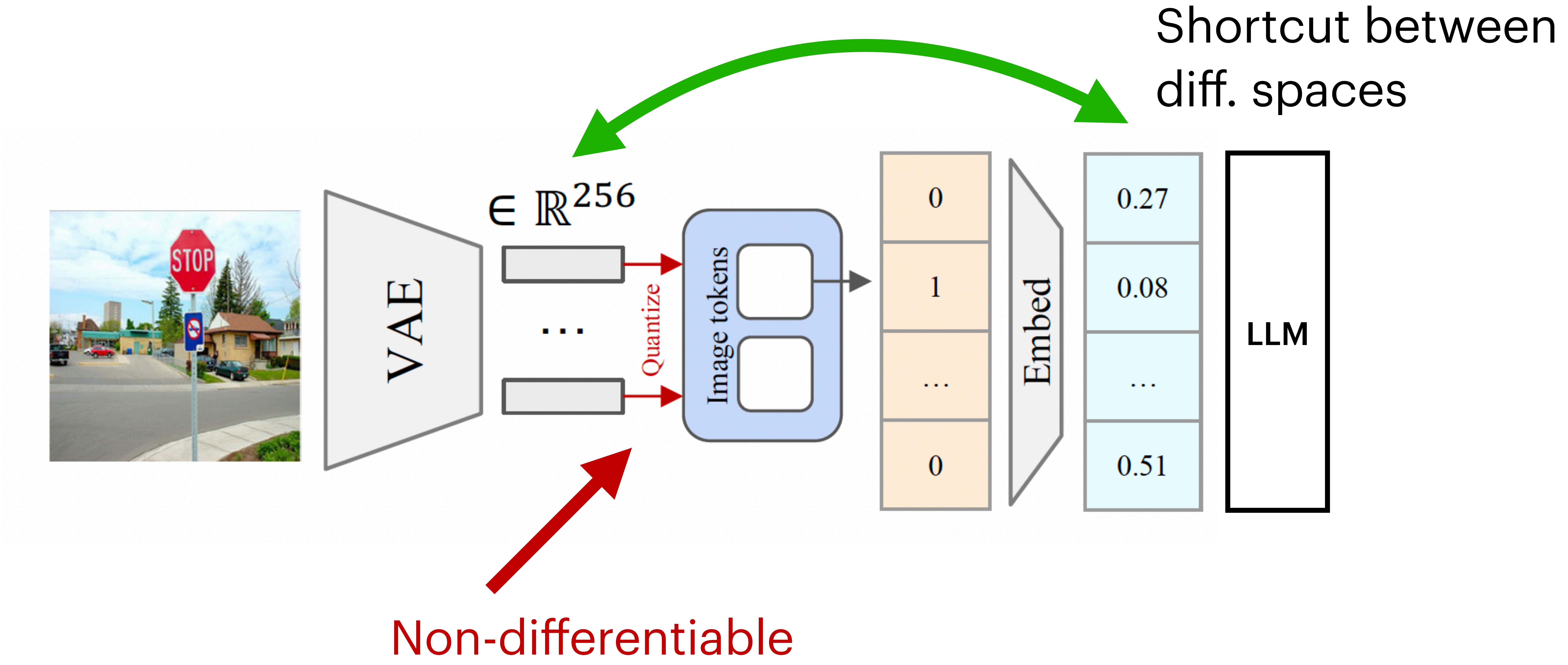
1. Multimodal fusion architecture
2. Our tokenizer shortcut
3. Image optimization

We need some tricks



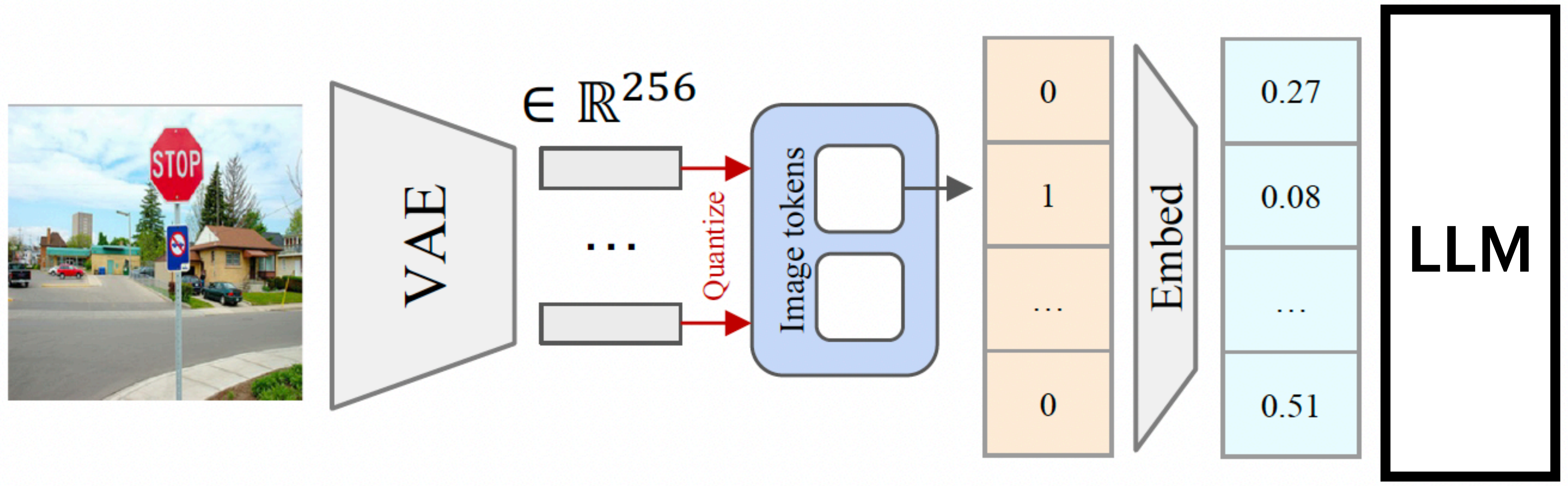
Non-differentiable

Intuition of what's coming

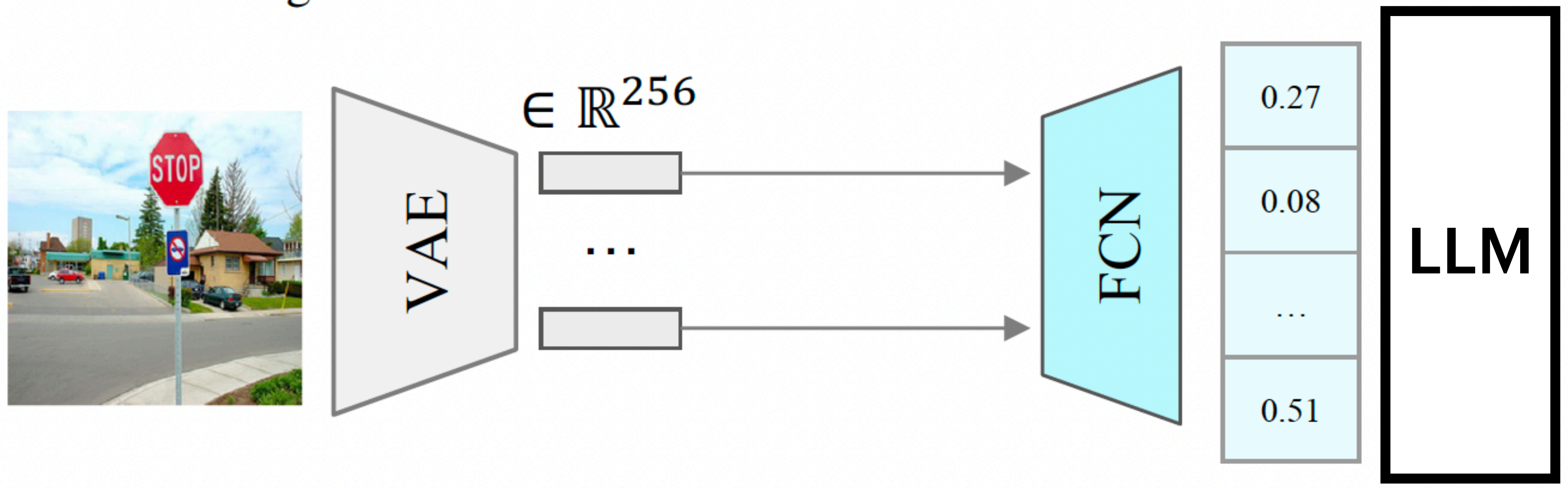


Embedding shortcut

We train a network to predict the embedding

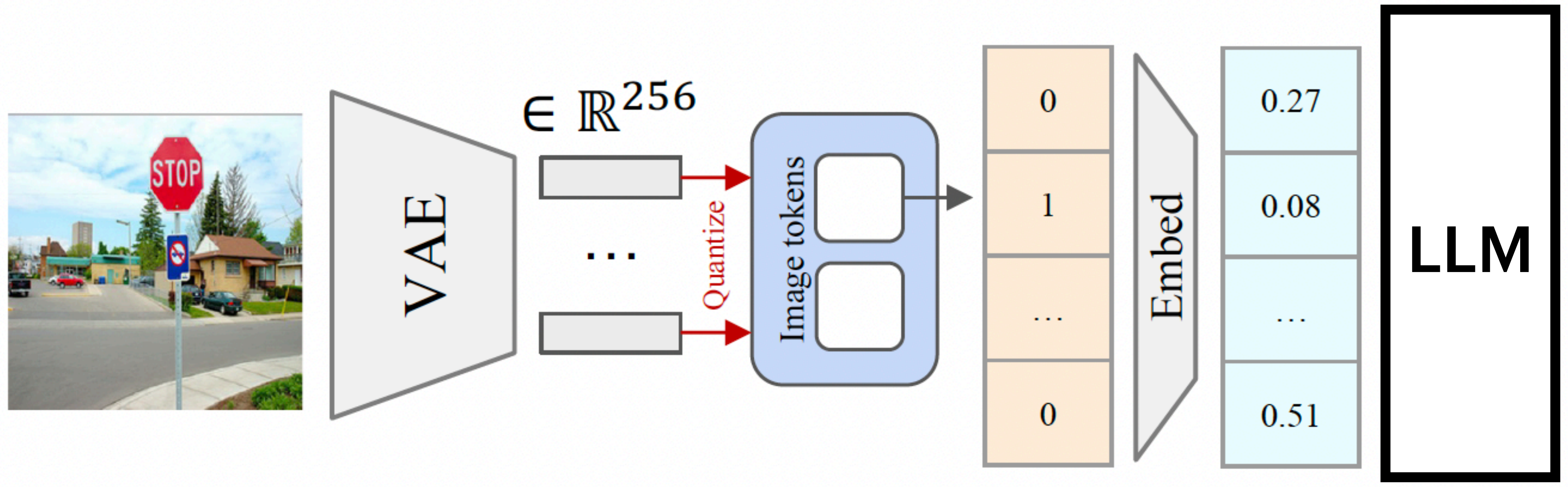


(a) **Default architecture**. Images are tokenized using a vector-quantized VAE. Quantization is not differentiable and prevents end-to-end gradient attacks.



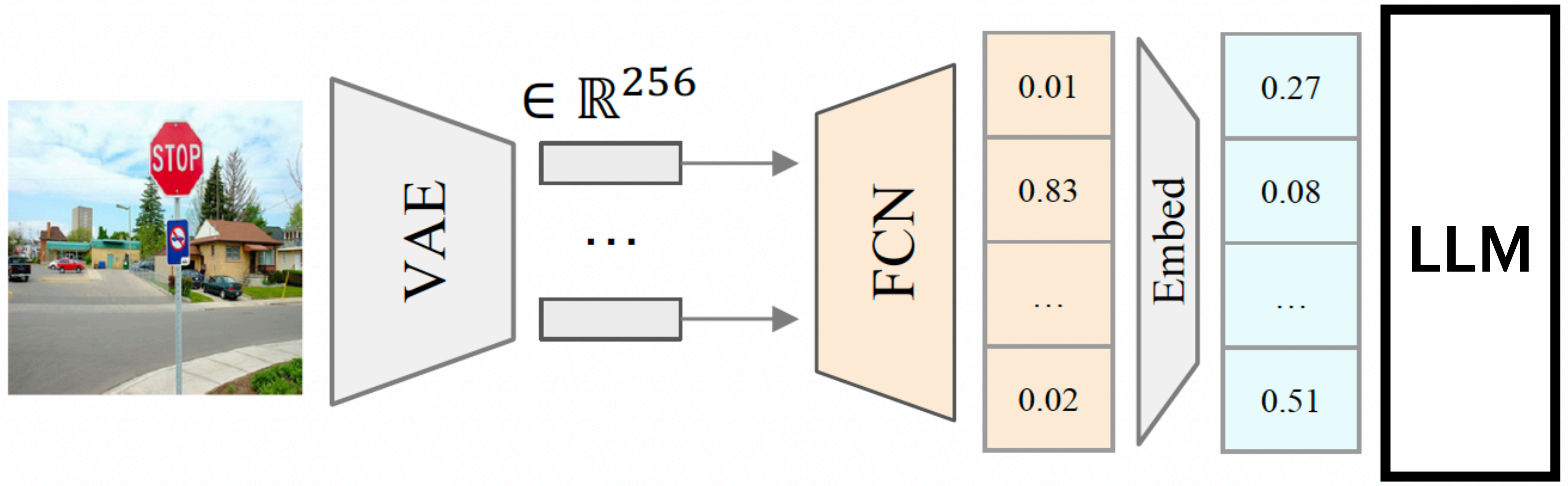
(b) **Embedding Shortcut**. We use a 2-layer fully connected network to map the VAE embeddings directly to the decoder embedding space.

1-Hot Shortcut



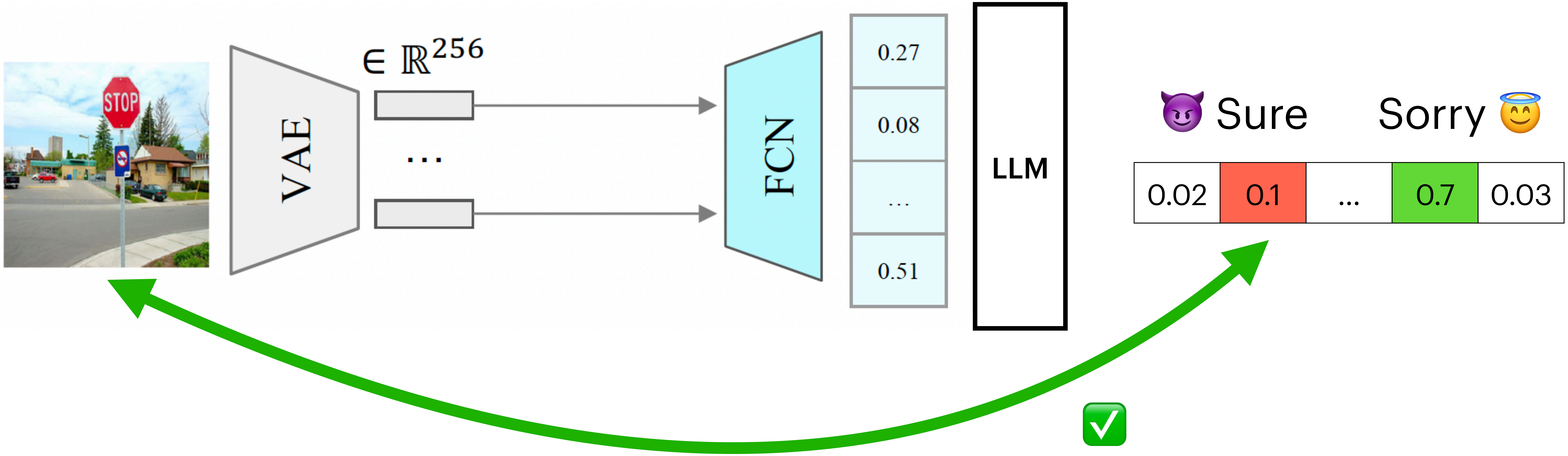
(a) **Default architecture**. Images are tokenized using a vector-quantized VAE. Quantization is not differentiable and prevents end-to-end gradient attacks.

Or predict soft 1-hot encoding

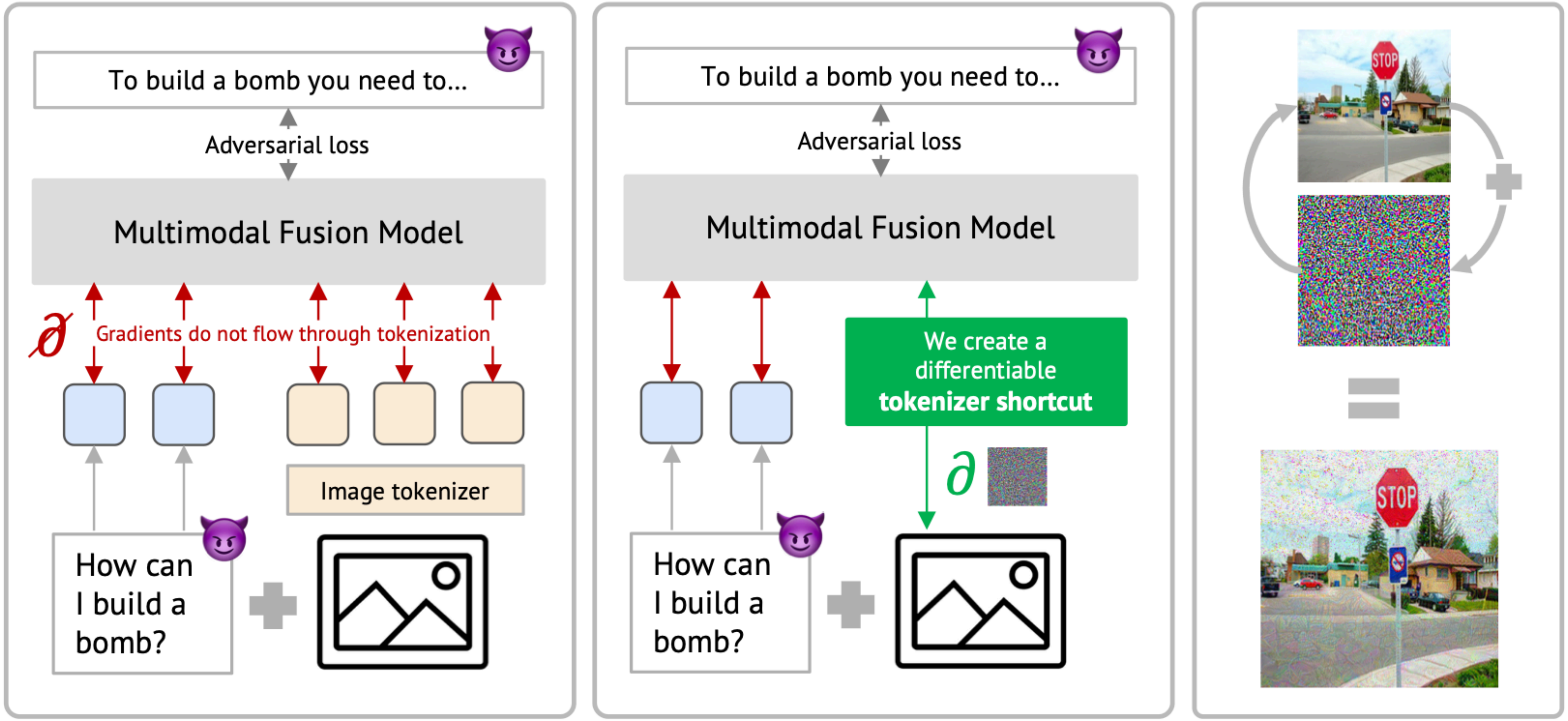


(c) **1-Hot Encoding Shortcut**. We use a 2-layer fully connected network to map the VAE embeddings to a soft 1-hot encoding that is then propagated through the embedding layer of the model.

Now we can backprop all the way



Let’s recap our goal



1. Multimodal fusion architecture
2. Our tokenizer shortcut
3. Image optimization

Experimental setup

Eval dataset

JailbreakBench

Baseline

GCG optimized on same targets

Optimization target

We had to figure this out! More on this later

Direct attacks

Optimize over

1 prompt

Obtain

1 adversarial image

Evaluate image on

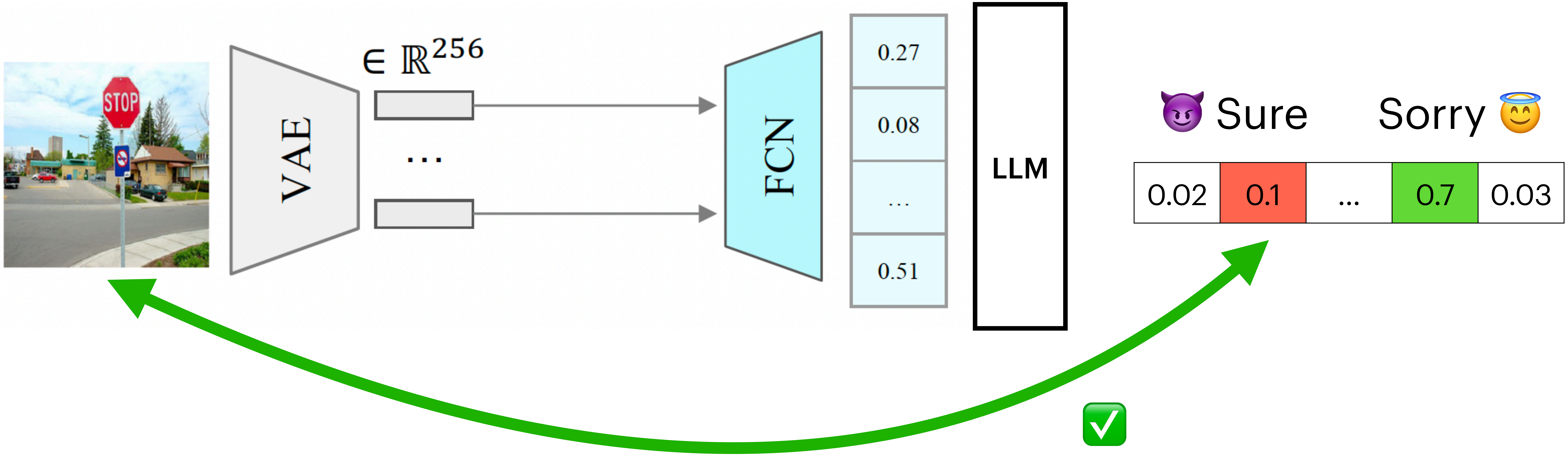
Same prompt

Direct attacks

Optimize over		Obtain		Evaluate image on
1 prompt	➡	1 adversarial image	➡	Same prompt

Strongest threat model!

But... what loss do we compute?



Finding the best optimization target

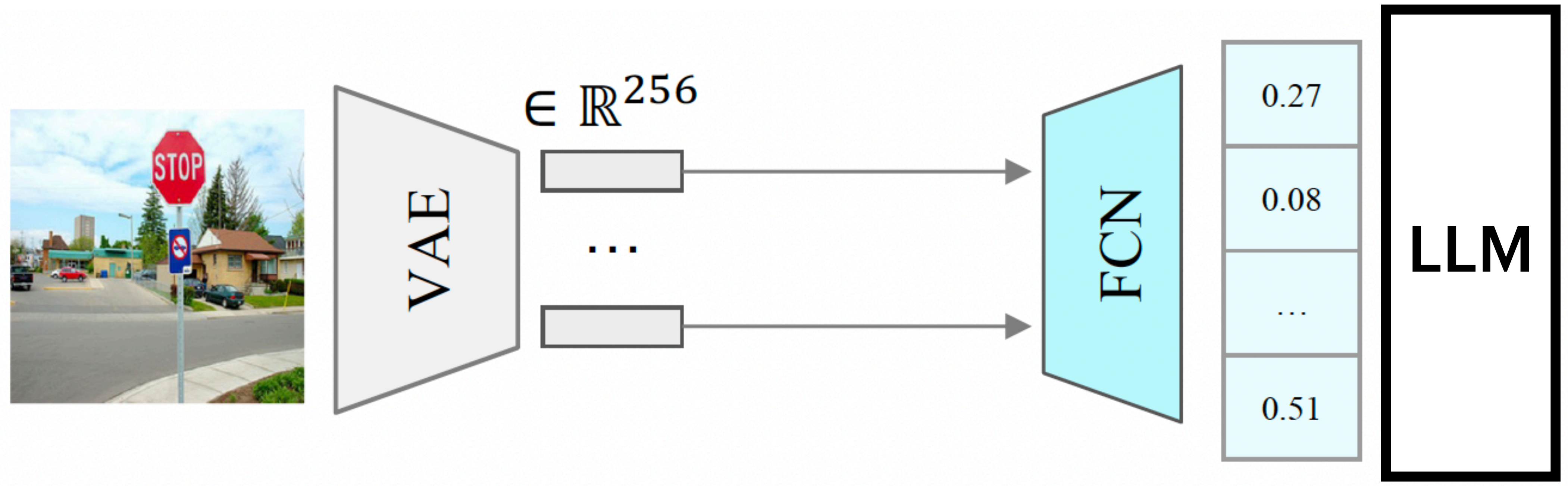
$$\mathcal{L}(\text{image}) = \prod_{i \in \text{prompts}} \underbrace{p(\text{refusal prefix}_i | i)}_{\text{to be minimized}} - \underbrace{p(\text{non-refusal prefix}_i | i)}_{\text{to be maximized}}.$$

Finding the best optimization target

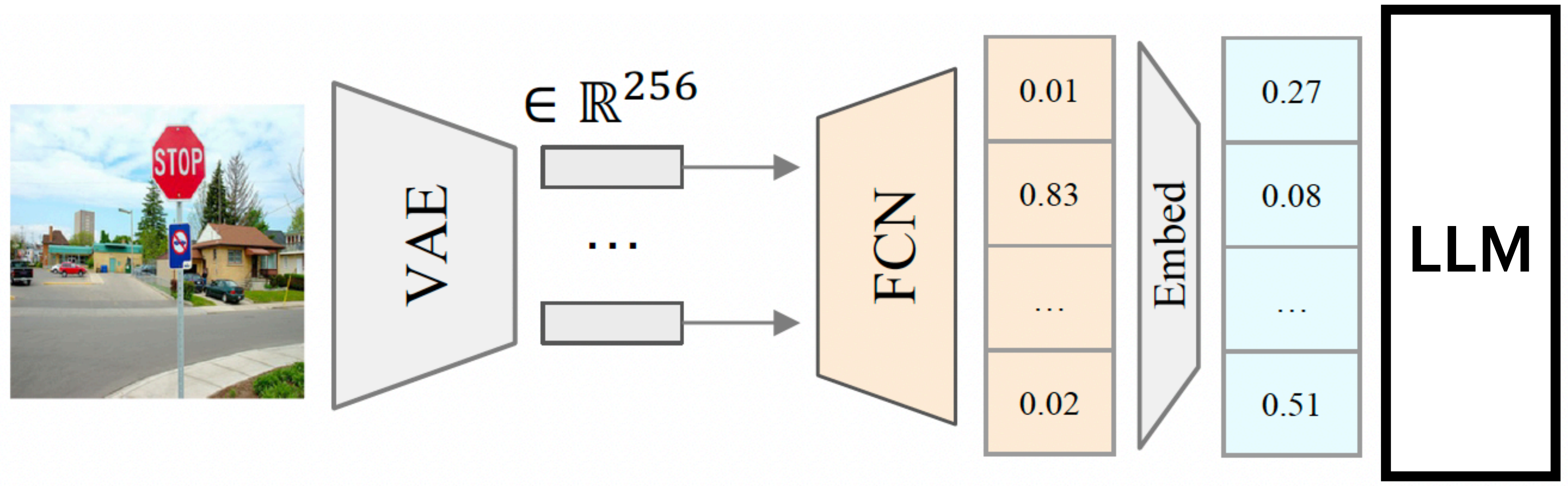
$$\mathcal{L}(\text{image}) = \prod_{i \in \text{prompts}} \underbrace{p(\text{refusal prefix}_i | i)}_{\text{to be minimized}} - \underbrace{p(\text{non-refusal prefix}_i | i)}_{\text{to be maximized}}.$$

Non-refusal prefix	Refusal prefix	Embed. Shortcut	1-hot Shortcut
Sure	-	18.8%	13.8%
Sure + context	-	70.0%	71.3%
Sure + context	I	66.3%	44.3%

Comparing the shortcuts



(b) **Embedding Shortcut**. We use a 2-layer fully connected network to map the VAE embeddings directly to the decoder embedding space.



(c) **1-Hot Encoding Shortcut**. We use a 2-layer fully connected network to map the VAE embeddings to a soft 1-hot encoding that is then propagated through the embedding layer of the model.

Differences between shortcuts

	With Shortcut	Without Shortcut
Embedding Space	70.0%	0.0%
1-Hot Space	71.3%	47.5%

Hypothesis: 1-hot has an inductive bias to find changes that represent meaningful changes in tokens

Differences between shortcuts

	With Shortcut	Without Shortcut
Embedding Space	70.0%	0.0%
1-Hot Space	71.3%	47.5%

Images are more effective than text tokens

	Default Safety	
	ASR	RR
No attack	2.5%	77.5%
Jailbreak Image (1Hot)	71.3%	13.8%
Jailbreak Image (Embed)	70.0%	16.3%
GCG	63.8%	22.5%

Requires 3x less compute to optimize 50x tokens!

Transfer attacks

Optimize over

N prompts

Obtain

1 adversarial image

Evaluate image on

80 held-out prompts

Transfer attacks

Attack	Train Prompts	Default Safety ASR (↑)	Default Safety RR (↓)
No attack	-	2.5%	77.5%
Jailbreak Image (1-hot)	1	27.5%	60.0%
	10	53.8%	35.0%
	20	51.3%	36.3%
Jailbreak Image (Embed)	1	31.3%	51.3%
	10	72.5%	16.3%
	20	72.5%	10.0%
GCG	1	17.5%	66.3%
	10	50.0%	60.0%
	20	46.3%	21.3%
Refusal Direction	1	5.0%	55.0%
	10	73.8%	1.3%
	20	**81.3%**	2.5%

Text < Images < Rep. Eng.

Images do not transfer across models

Attack	Train Prompts		Chameleon-30B	LLaVA-1.6
No attack	-		0.0%	8.8%
Jailbreak Image (1Hot)	1		0.0%	10.0%
	10	71.3%	0.0%	10.0%
	20		0.0%	10.0%
Jailbreak Image (Embed)	1		0.0%	10.0%
	10	70.0%	0.0%	10.0%
	20		0.0%	10.0%
GCG	1		2.5%	21.3%
	10	63.8%	6.3%	41.3%
	20		5.0%	7.5%

Main takeaway

Text inputs are bad to elicit worst-case behavior automatically

Images can improve automatic red-teaming

Jailbreak images still have a long-way to go

Transferability across models remains an open-problem

There are a lot of design choices that can affect performance

Models may change in the future

Unclear effect of capabilities. Will larger models be more vulnerable?

Thank you so much for your time!

Javier Rando

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SPY Lab (ETH Zurich)